

LOW-LEVEL AND HIGH-LEVEL CONTROL OF OVER-ACTUATED VEHICLES

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ABSTRACT

As the world is shifting towards automotive electrification in their goal towards a greener planet, another movement that has been growing drastically over the years has been the growth of Connected and Automated Vehicles. However, in order to ensure the best energy efficiency towards this, a study has been conducted on obtaining the best energy savings from the vehicle at different levels. In this paper, a focus on the energy savings at the Lower powertrain level due to the usage of in-wheel motors giving rise to the over-actuated vehicle is compared against the traditional Centralized powertrain structure for the Olli Shuttle by Local Motors. This work has also been validated by means of experiments with the Olli Shuttle. Also, another focus in the paper has been on the Higher-Level Decision Controller which is also being presented here where an optimum drive-cycle, obtained by means of optimizing the velocity of a baseline drive-cycle, is being provided as inputs to the Lower powertrain level which is then eventually optimized at the lower level for best energy efficiency. The energy consumption due to the optimization at the higher level is also being compared in this report.

ACKNOWLEDGEMENTS

Firstly, I would like to express my deepest gratitude to my advisor, Dr. Yan Chen for letting me be a part of this prestigious project and for providing his guidance and invaluable inputs in every step along the way and also for introducing me to the field of Automotive controls. Also, sincere thanks to the students of Dynamic Systems and Control Laboratory for helping me with their technical expertise. Furthermore, Special thanks to my colleagues Fengchen Wang and Tulja Lal Birju for lending me their support in this endeavor.

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ABBREVIATIONS

IWM = In-Wheel Motors

CM = Central Motors

TTTL = Time to Traffic Light

TL = Traffic Light

D_{TLx} = Distance to Traffic Light x

SP = Signal Phase

WT = Waiting Time

AT = Arrival Time

RP = Red Phase

CHAPTER-1

INTRODUCTION

Connected and Automated Vehicles (CAVs) represents one of the biggest trends and opportunities in the automotive industry. CAVs reduce the human factor which is the primary reason for almost 90% of all the accidents and deaths occurring over the world^[1]. Therefore, CAVs have the potential for great growth in the automobile industry. However, despite the potential of the technology, only a limited amount of attention has been focused on reducing the energy consumption due to CAVs. There are some studies that suggest that the energy consumption change in the transportation sector ranges between -60% to +200%^[2]. One of the primary objectives of this project is to narrow down the energy consumption range for a CAV named Olli being developed by Local Motors.



Fig 1.1. Olli Shuttle by Local Motors

The Olli is primarily being developed as a shuttle by Local Motors intended to transport passengers with a capacity of transporting up to 10 passengers. It deploys connection and automation techniques with a traditional Centralized powertrain structure as a normal Electric Vehicle. However, with the advancement of the new in-wheel motors which negates the need for a powertrain clubbing all the transmission units in a single wheel hub, it can be seen that in-wheel motors (IWMs) provide great potential in electric vehicles to simplify the configuration of power transmission, such that the energy cost is reduced.

Low Level Control:

The main idea of this project is to utilize distributed redundant actuation to increase system reliability, achieve system reconfigurability, and extend system performance. Redundant

actuators, mathematically corresponding to extra solutions, were adopted as an approach to satisfy the increasing demands on safety and performance in large and complicated engineering systems ^[11]. Many existing physical systems, such as marine vessels ^[5], airplanes ^[6], and ground vehicles ^[7], can be classified as over-actuated (OA) systems, in which the number of actuators is greater than the control degrees of freedom. Specifically, electric vehicles can be constructed as OA systems by utilizing IWM technology ^[8].

IWM is a new and advanced actuation technology, which integrates an electric motor, a gear reduction module, and a braking system within one wheel and consequently changes the traditional centralized powertrain into a distributed actuation ^[9]. With the help of independently and directly actuated IWMs, many novel functions can be achieved by electric vehicles, compared with the traditionally central powertrain structure.

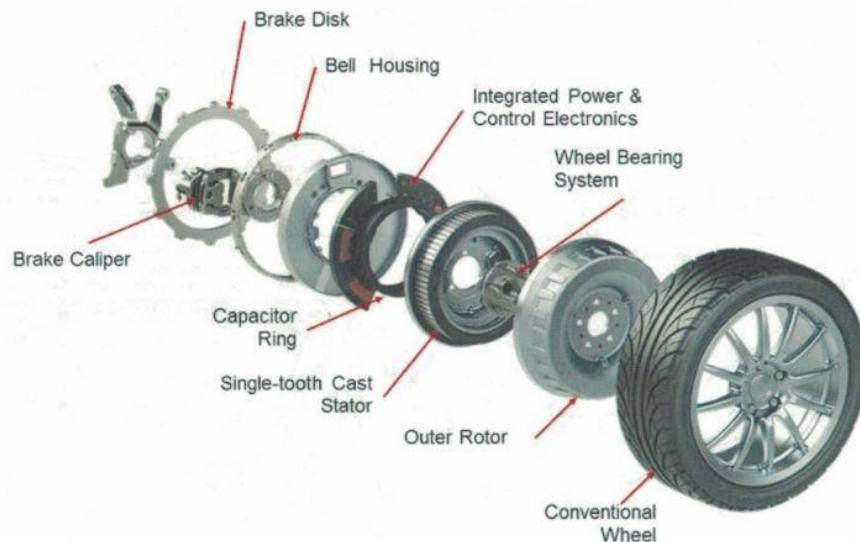


Figure. 1.2 Protean In-Wheel Motor ^[16]

For example, electric differentials can be easily and efficiently implemented for an OA electric vehicle in different cornering and turning maneuvers ^[14].

Different from the traditional powertrain by utilizing complicated and specially designed mechanical differentials to achieve torque vectoring on inner and outer wheels, the electric differentials enabled by in-wheel EMs technology were directly controlled by the actuated motor torques according to different rotational speed references of inner and outer wheels. The advantages brought by electric differentials are not only higher efficiency without energy loss on mechanical connections, but also safer operation due to the fast and accurate torque control

of in-wheel EMs. Motivated by the advantages of in-wheel EMs technologies, an over-actuated CAV, independently actuated by two in-wheel EMs is proposed for this project. The novel powertrain architecture will enhance the control performance and energy efficiency of the Olli vehicle, compared to the baseline Olli vehicle performance, but especially when compared with the traditional centralized powertrain structure. A novel on-board controller that will make the required low-level powertrain decisions will be developed.



Figure. 1.3. EMRAX228 Central Motor^[17]

This low-level powertrain controller (LLPC) will control the various powertrain components, including the EMs and ESS to provide the required propulsion, braking, and steering actuation while maximizing the energy efficiency of the vehicle operation. The main challenges of such the proposed vehicle is how to fully utilize the advantages of redundant actuations to achieve estimation, control, and optimization of the whole vehicle, synergizing with sensing fusion and motion planning systems. Although various sensing, modeling, and control are discussed for automated driving, few research efforts have focused on the energy optimization of automated driving vehicles. Since the Olli vehicle is a pure EV, energy efficiency optimization is a key step to extend the driving range. Moreover, the characteristics of OA with two in-wheel EMs and two steering-by-wire systems provide great potentials to achieve more energy savings, compared with the centralized electric powertrain.

For Phase-1 of this project, only the longitudinal dynamics of the Olli is being considered here where the Torque is being distributed in half between the 2 in-wheel motors. Here, two in-wheel motors are the new suggested method for Olli however the idea behind the use of four

in-wheel motors is also being briefly discussed here. Phase-2 of this project will be factoring in the lateral dynamics and yaw dynamics to optimize the Torque distribution.

High Level Control:

Similar to what's happening with the on-board novel Low-level controller, the high-level decisions are also made by the controller which determines the route the Olli Shuttle is going to take providing it as inputs to the Low-Level controller. This high-level decision controller (HLDC) will interface externally with surrounding vehicles and infrastructure to determine the optimized propulsion, braking, and steering requirements. The HLDC will also interface internally with the Olli's low-level powertrain controller (LLPC). The High-level controller here gives velocity inputs to the low-level controller which performs energy optimization at the lower level for overall reduced energy consumption. New communication protocols and optimization algorithms will be developed to dynamically control the propulsion, braking, and steering decisions of the CAV based on traffic and road topography information. The Olli shuttle bus uses connectivity for vehicle to everything (V2X) communications including look-ahead information of road topography and signal phase and timing (Spat) information, which can allow for optimized propulsion, braking, and steering actuation of the automated driving system (ADS).

Specifically, the HLDC will obtain traffic information such as prevailing traffic speed, congestion and work zones, traffic SPaT information, and incident information from the traffic management center through V2I communication. The HLDC will also obtain road topography (grade) information from the traffic management center, when available.

CHAPTER-2

LOW LEVEL SIMULATION, VALIDATION AND RESULTS

LOW LEVEL SIMULATION:

Here, we analyze the energy efficiency performance of electric vehicle driven by one central electric motor or two in-wheel motors, which are both utilized by Olli. Two cases are investigated: Olli driven by one central electric motor (Emrax 228) and Olli driven by two IWMs (Pd18). The efficiency maps of Emrax 228 and Pd18 are shown in 2.1. The braking efficiency property of Emrax 228 motor is absent, and hence only the driving efficiency performance is discussed in the following.

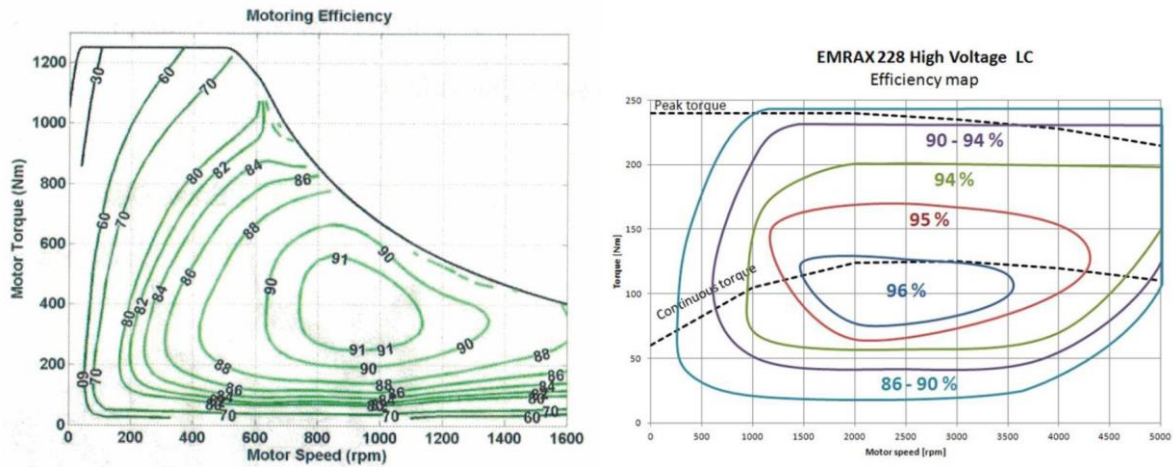


Fig. 2.1. Efficiency maps of the central motor and IWM (from motor specifics).

For Olli with the central motor, it has a transmission gear ratio as 9.59. Moreover, the transmission, differential, and ball-joint efficiency reduction of Olli with the single central motor is also considered, which is 0.866 in total [1]. Comparison of the efficiency maps for central motor (Emrax228) and IWM (PD18) is presented in Fig. 2. Table 1 shows the parameters that have been included in the simulation that has also been provided by the Local Motors.

Table 2.1. Parameters used in simulation.

Parameters	Values
Vehicle mass including passengers	3266 kg
Gearbox transmission ratio	9.59
Wheel diameter	0.4572 m

Front cross section area	4.2025 m ²
Drag coefficient	0.54
Rolling resistance coefficient	0.015
Maximum acceleration	1.5 m/s ²
Maximum driving speed	35 mph

Within Fig. 2, the operation region labeled by red dot area means that the powertrain configuration with two in-wheel motors always has higher energy efficiency than the one with the single central motor. Therefore, the operation region should be used to provide the guidance to generate the Olli driving maneuvers. Note that the operation region defined in Fig. 2 actually covers some normal operations of Olli. It is obvious from the operating region that the acceleration cannot be greater than 1 m/s² for the in-wheel motor to perform efficiently than the central motor. Moreover, it is also seen that during most of the constant speed motions, the in-wheel motor is going to perform efficiently than central motor.

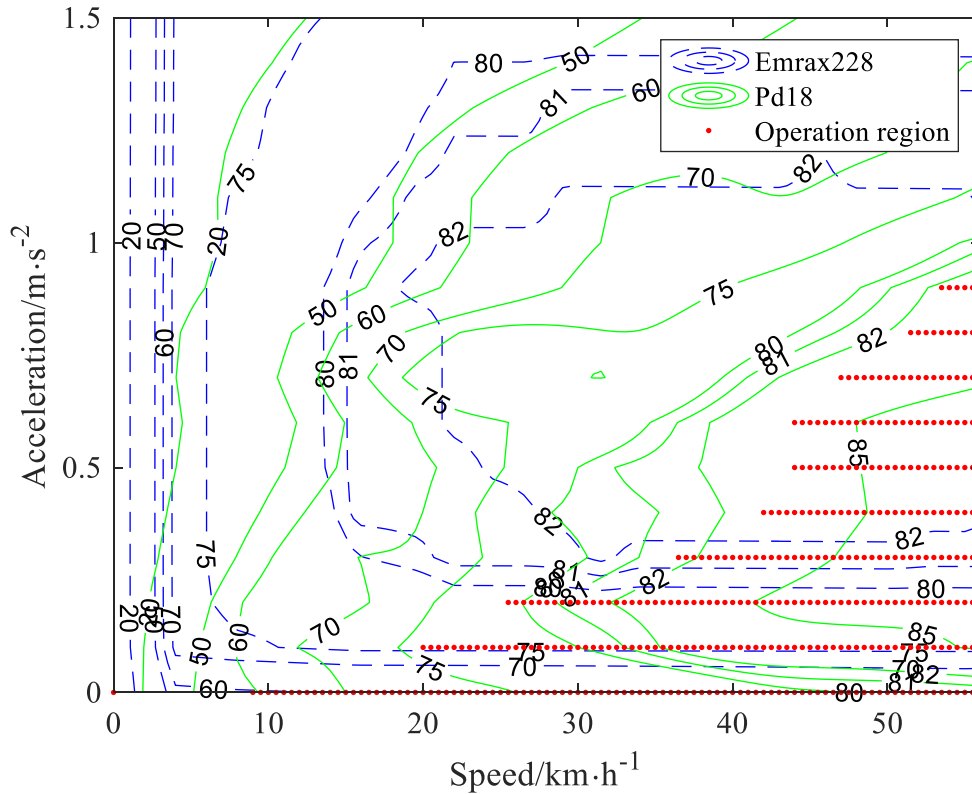


Fig. 2.2. Efficiency maps and the proposed operation region.

The typical Olli driving maneuver has been designed using the operating region in Fig 2.2 as a reference. The different maneuvers are explained below case by case.

1. Olli at a high constant speed (40 km/h = 25 mph)

The simulation results are shown below. In Fig. 2.1.1, the vehicle speeds for the central motor case and the in-wheel motors case are both able to track the reference speed at 40 km/h.

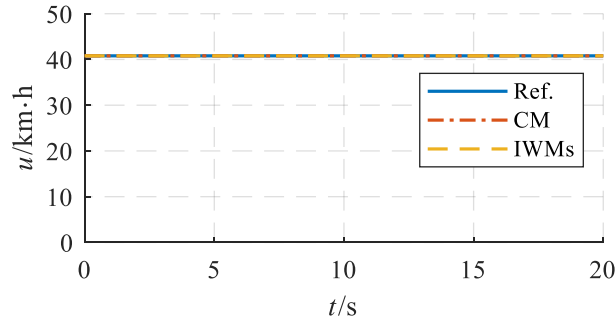


Fig. 2.1.1. Driving speed profile and tracking performances.

In Fig. 2.1.2 and Fig. 2.1.3, the total required power and force are determined by using the specifications of Olli, which are expected to overcome the driving resistance. Considering the mechanical configuration, Fig. 2.1.4 and Fig. 2.1.5 show the required torque and rotation speed of each motor in both cases. Compared with the central motor case, it is reasonable to observe the lower motor rotation speed and higher required motor torque for the in-wheel motor, owing to the absence of the transmission gearbox.

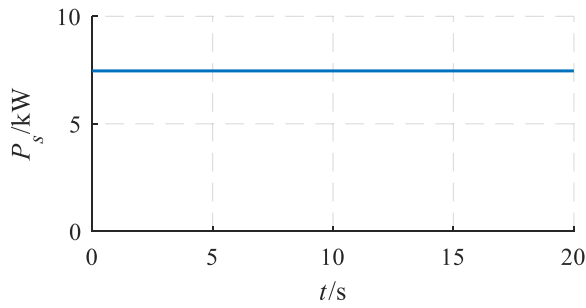


Fig. 2.1.2. Required total power.

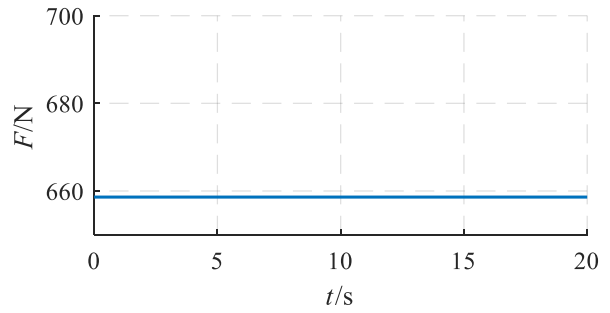


Fig. 2.1.3. Required total force.

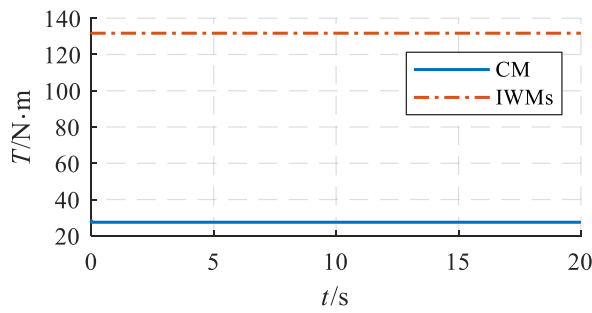


Fig. 2.1.4. Required motor torque.

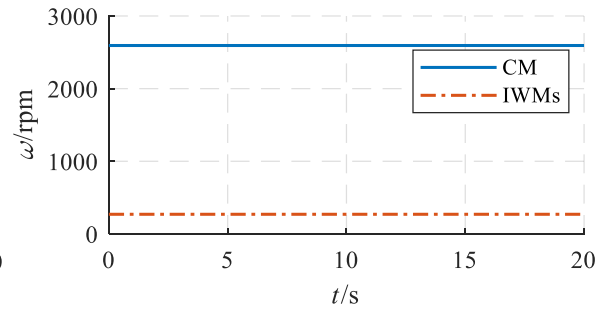


Fig. 2.1.5. Required motor rotation speed.

The energy efficiency and power consumptions results are presented in Fig. 2.1.6. and Fig. 2.1.7. It can be seen that the constant speed of 40 km/h (cruising speed) lies in the operation region defined in Fig. 2.2. Based on the efficiency map and the operation region, the efficiency of the in-wheel motors is higher than that of the central motor for the same condition as the maneuver here lies in the red dotted area within Fig 2.2. Therefore, the Olli with in-wheel motors is more efficient than the one with the central motor at the constant cruising speed with less energy consumption.

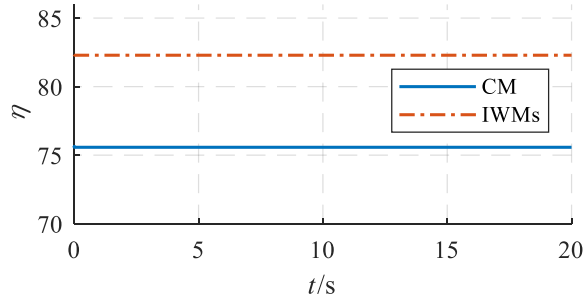


Fig. 2.1.6. Energy efficiency of motors.

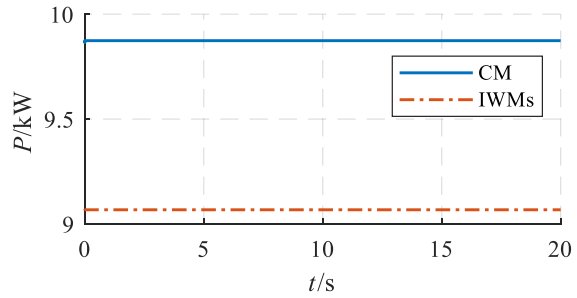


Fig. 2.1.7. Consumed power of motors.

Overall, it is observed that, when the Olli is running at the constant cruising speed around 40 km/h (25 mph), the IWM driven efficiency is much higher than that of the central motor for this common Olli driving maneuvers and it also consumes less power. **(About 8.8% power saving).**

2. Olli at a low constant speed (25 km/h = 15 mph)

The simulation results when the speed is constant at 25 km/h are shown. In Fig. 2.2.1, the speeds for the central motor case and in-wheel motors case are both able to track the reference speed at 25 km/h.

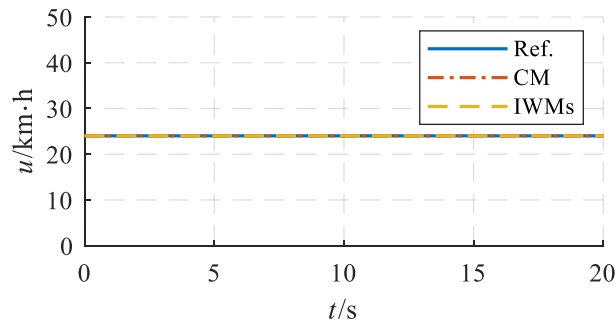


Fig. 2.2.1. Driving speed profile and tracking performances.

In Fig. 2.2.2 and Fig. 2.2.3, the total required power and force are determined by using the specifications of Olli, which are expected to overcome the driving resistance. Considering the mechanical configuration, Fig. 2.2.4 and Fig. 2.2.5 show the required torque and rotation speed of each motor in both cases. Compared with the central motor case, it is reasonable to observe the lower motor rotation speed and higher required motor torque for the in-wheel motor, owing to the absence of the transmission gearbox.

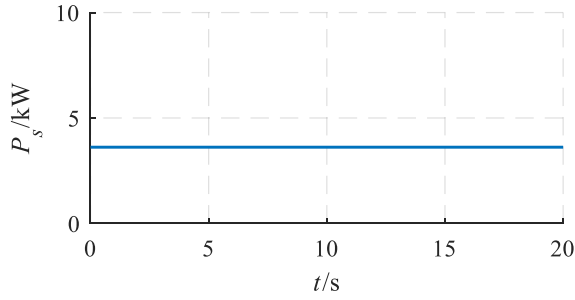


Fig. 2.2.2. Required total power.

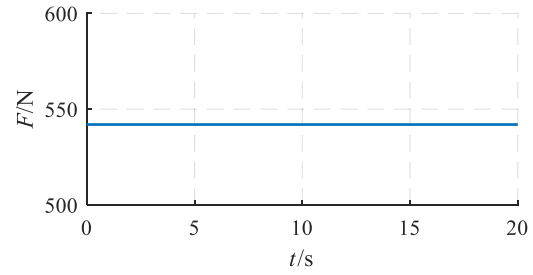


Fig. 2.2.3. Required total force.

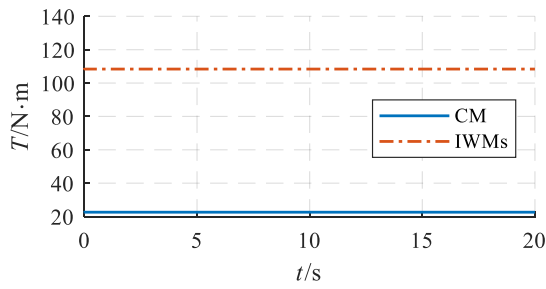


Fig. 2.2.4. Required motor torque.

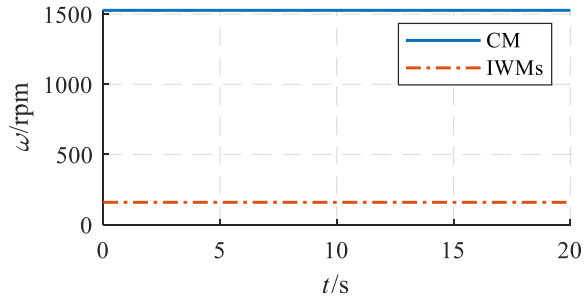


Fig. 2.2.5. Required motor rotation speed.

The energy efficiency results are presented in Fig. 2.2.6 and Fig. 2.2.7. It can be seen that the constant speed of 40 km/h (cruising speed) lies in the operation region defined in Fig. 2.2. Based on the efficiency map and the operation region, the efficiency of the in-wheel motors is higher than that of the central motor for the same condition as the maneuver here lies in the red dotted area within Fig 2.2. Therefore, the Olli with in-wheel motors is more efficient than the one with the central motor at the given low speed with less energy consumption.

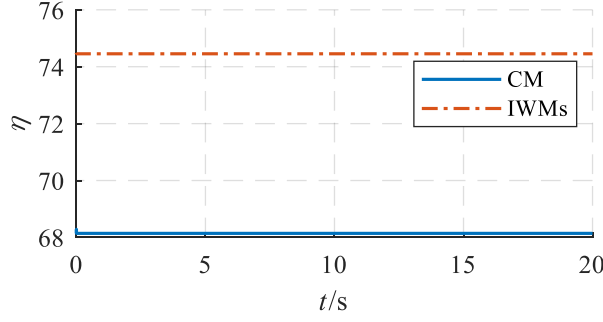


Fig. 2.2.6. Energy efficiency of motors.

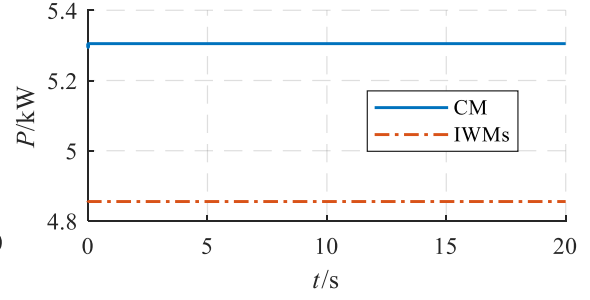


Fig. 2.2.7. Consumed power of motors.

Overall, it is observed that, when the Olli is running at a constant low speed around 25 km/h (15 mph), the IWM driven efficiency is much higher than that of the central motor for this common Olli driving maneuvers and it also consumes less power. **(About 9.3% power saving).**

3. Olli accelerating from a medium constant speed to the max speed

The simulation results when the Olli is accelerating ($a=0.2 \text{ m/s}^2$) from the cruising speed of 35 km/h to the maximum speed of 45 km/h are shown. In Fig. .1, the speeds of the central motor case and in-wheel motors case are able to track the reference.

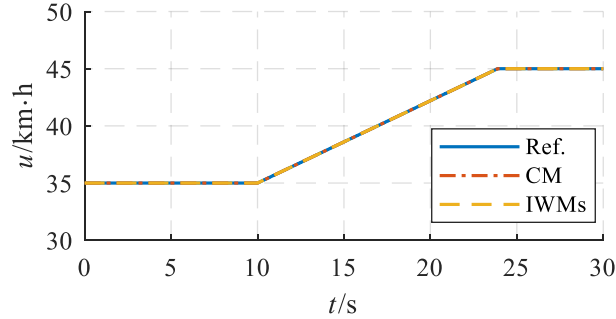


Fig. 2.3.1. Driving speed profile and tracking performance.

In Fig. 2.3.2. and Fig. 2.3.3., the total required power and force are determined by using the specifications of Olli, which are expected to overcome the driving resistance. Considering the mechanical configuration, Fig. 2.3.4. and Fig. 2.3.5. show the required torque and rotation speed of each motor in both cases.

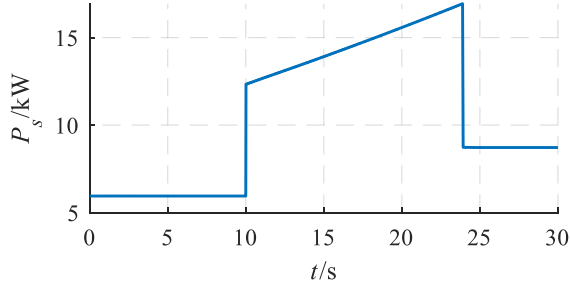


Fig. 2.3.2. Required total power.

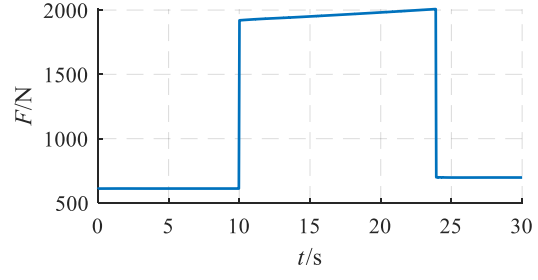


Fig. 2.3.3. Required total force.

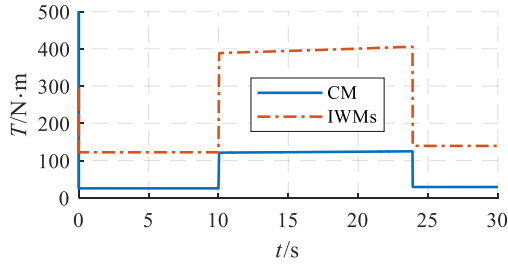


Fig. 2.3.4. Required motor torque.

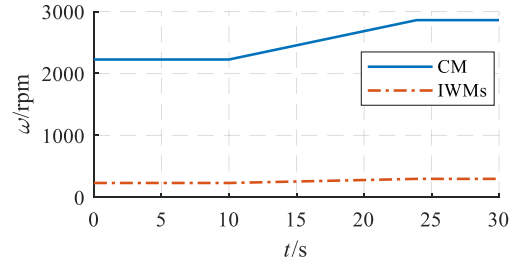


Fig. 2.3.5. Required motor rotation speed.

The energy efficiency results are presented in Fig. 2.3.6. and Fig. 2.3.7. It can be seen that the speed of 35 kmph and speed of 45 kmph lies in the operation region defined in Fig. 2.2. Based on the efficiency map and the operation region, the efficiency of the in-wheel motors is higher than that of the central motor for the same condition as the maneuver here lies in the red dotted area within Fig 2.2.

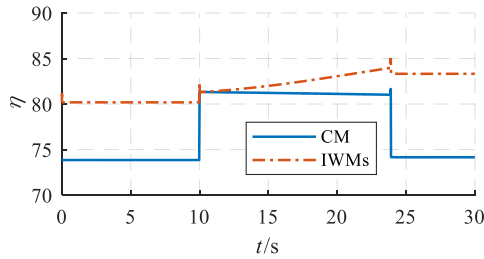


Fig. 2.3.6 Energy efficiency of motors.

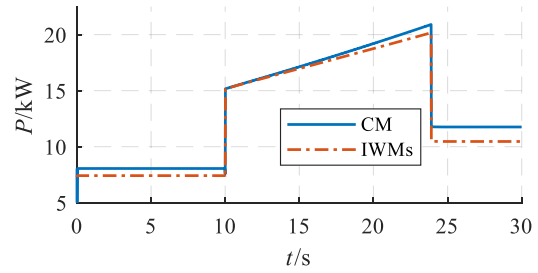


Fig. 2.3.7 Consumed power of motors.

From Fig. 2.3.6. and Fig. 2.3.7, it is evident that it stands true to all the previous observations that the in-wheel motor performs has better efficiency and consumes less power when at constant speed. Also, if the acceleration is low at high speeds, the IWM performs better in comparison with the central motor. **(About 10.72% power saving).**

From the above results, we can draw a conclusion that in-wheel motor can provide better energy efficiency than central motor under the most common maneuvers. Three typical driving

maneuvers give about **9% ~ 10%** power saving. These results have also been validated experimentally.

Additionally, all the above simulation results have been obtained considering a zero-degree gradient. However, when we consider a gradient, the operating region is going to change. The below figure shows the operating region when the slope is 2 degrees and 9 degrees respectively.

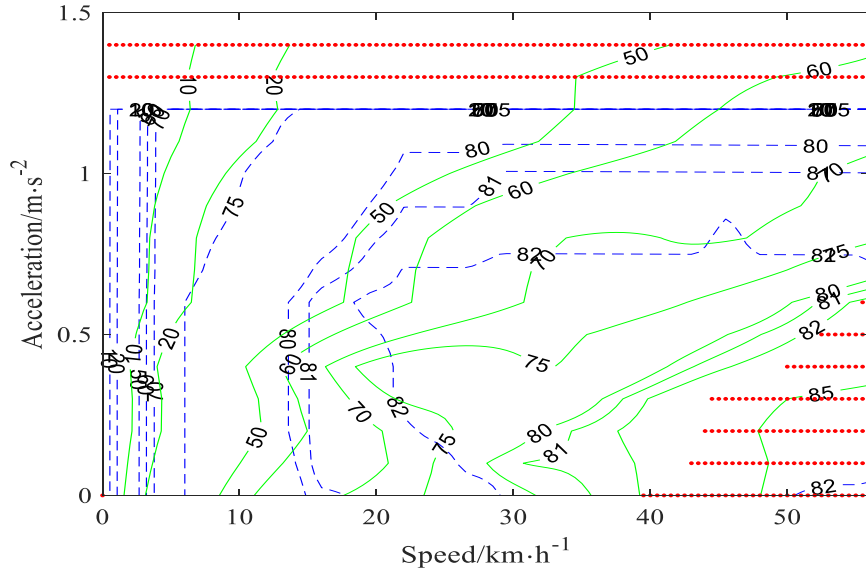


Fig.2.3. Operation region at 2 deg slope.

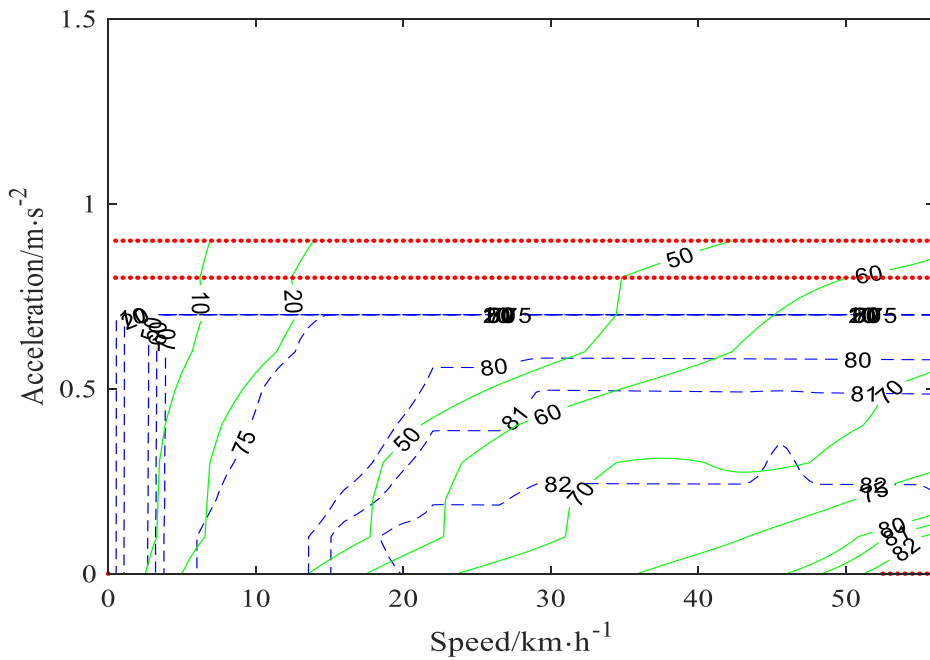


Fig. 2.4. Operation region at 9 deg slope.

EXPERIMENTAL VALIDATION & RUN DATA ANALYSIS:

The above simulations results were validated by means of experiments at the Local Motors Chandler facility. The experiments were conducted keeping in mind the assumptions and the constraints that were mentioned in the Design of Experiment (DOE) Statement. The experimental run data obtained were then analyzed for closeness with the simulation results for validation. As part of the DOE were certain controlled and uncontrollable parameters which were the constraints for the experiment.

Table 2.2: Controlled and Uncontrollable Parameters for the Experiment

Controlled Parameters	Uncontrollable Parameters
- Vehicle velocity(constant speed maneuvers) - Battery SOC - Straight line testing - Same load	- Gradient of Terrain (expectedly zero) - Weather conditions - Top Speed of IWM (only touching top speed of 24 mph). CM also only tested up to 24 mph to achieve same velocity comparison.

For constant speed of 24 mph:

Because of the comparatively high-speed testing and the testing track length, only 6 seconds of the constant speed could be sustained for analysis. Therefore, these 6 seconds were the focal point for the analysis and the consequent comparison with the simulation results.

The below figures Fig 2.4.1 and Fig 2.4.2 shows the experimental results for vehicle velocity and the motor speed for the Central Motor and the in-wheel motor driven Olli respectively.

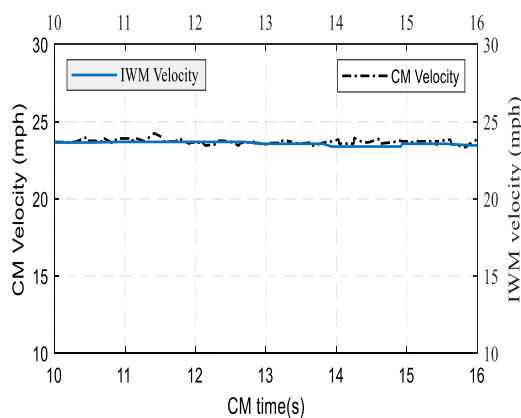


Fig 2.4.1: vehicle velocity comparison

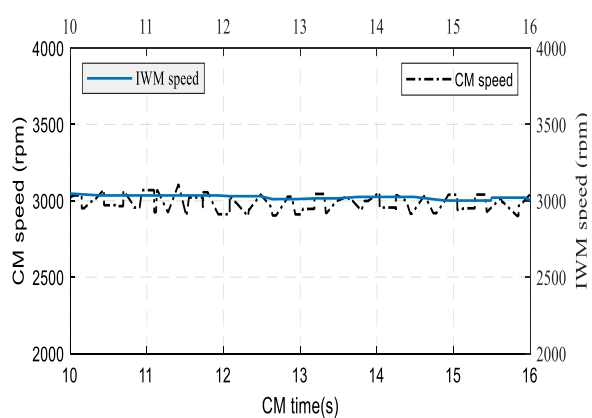


Fig 2.4.2: motor speed comparison

Fig 2.4.1 shows similar velocity for both the Ollis whereas the Fig 2.4.2 is also showing matching with the motor speed criterion comparison for CM and IWM where the Motor speed of CM is equivalent to 9.59 times the Motor speed of IWM ($\text{MotorSpeed}_{\text{CM}} = 9.59 * \text{MotorSpeed}_{\text{IWM}}$). These results show agreement with the simulation results.

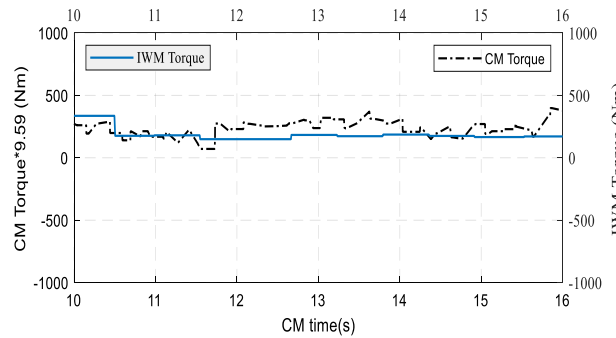


Fig 2.4.3: Motor Torque comparison

Fig 2.4.3 shows the comparison between the experimental motor torque for the CM and 2IWM. The Torque readings for the run data were obtained from the Accelerator Pedal Input and the forces acting. The above figure shows matching with the Torque criterion when 2 IWMs were compared against one CM where 2 times the motor torque of one IWM is equivalent to 9.59 times the motor torque of the one CM ($2 * \text{Torque}_{\text{IWM}} = 9.59 * \text{Torque}_{\text{CM}}$). The average Torque observed were 169 and 15 N-m for IWM and CM respectively. These results also show agreement with the simulation results.

In order to ensure complete accuracy of the numbers in the run data, the motor torque should be proportional to the battery current. Fig 2.4.4 and Fig 2.4.5 shows the necessary proportionality between the Motor torque and the current.

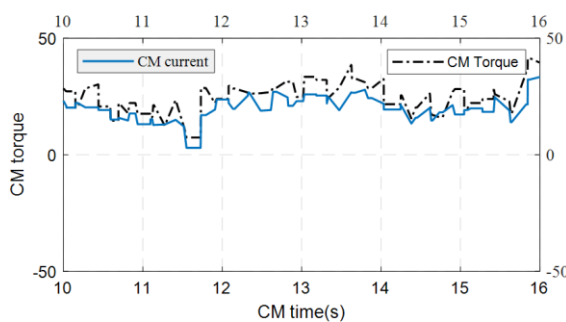


Fig 2.4.4: CM Torque & Current comparison

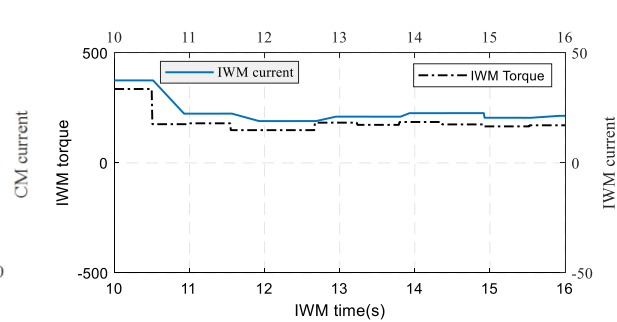


Fig 2.4.5: IWM Torque & Current Comparison

Both of the above figures shows proportionality between the Motor torque and current for CM and IWM except for some delays observed due to data acquisition. The average Battery current observed were 22A and 19A for IWM and CM respectively.

The key comparison is the comparison between the mechanical output power and the input power consumption. Fig 2.4.6, Fig 2.4.7 and Fig 2.4.8 shows the comparison between the output power, input power consumption and the efficiency comparison between the CM and the IWM after some polishing of the experimental run data.

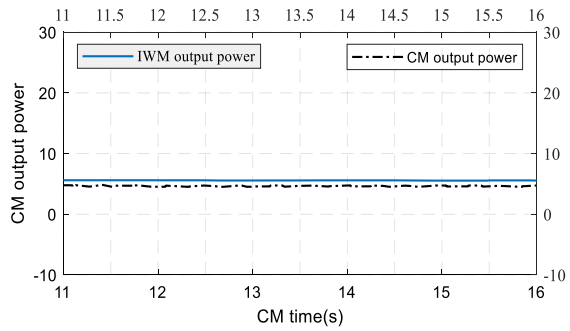


Fig 2.4.6: Output power comparison

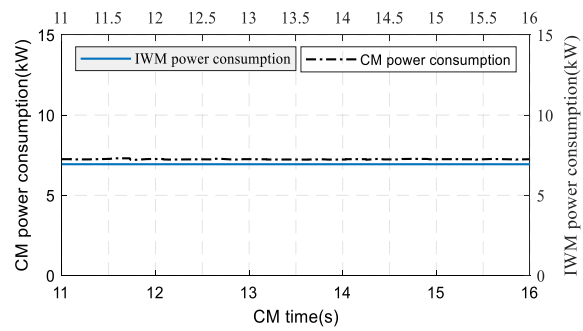


Fig 2.4.7: Power consumption comparison

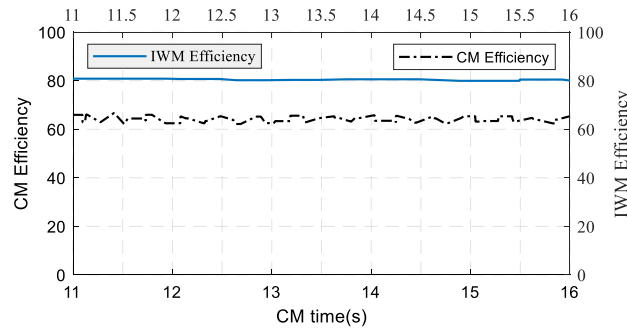


Fig 2.4.8: Efficiency comparison

The above plots for mechanical output power has been calculated by utilizing the average motor torque values and the input power consumption has been calculated by utilizing the average current. And, the efficiency has also been calculated by utilizing the mechanical power and the input electrical power. These results show agreement with the simulation results, ie, For similar output power & 24 mph constant speed, IWM shows less power consumed than CM (Avg Torque of 169 N-m and 15 N-m for IWM and CM respectively and Avg Battery current of 22A and 19A for IWM and CM respectively) with an observed **power savings of around 7.5 %** and for the efficiency, Average Efficiency_{IWM}(around 80%) greater than Average Efficiency_{CM}(around 65%) for 24 mph constant speed.

The Torque and motor speed of the two motors when plotted against the efficiency maps obtained from the product data sheets should also show the similar results obtained from the experimental run data.

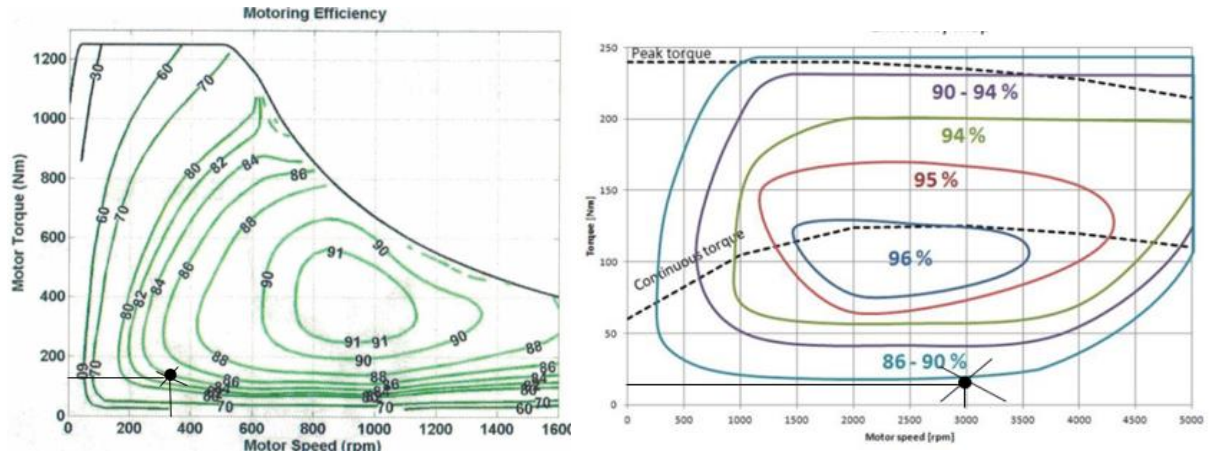


Fig 2.5: Efficiency of IWM and CM respectively against the output torque and motor speed as observed in the product datasheet

According to datasheets, for 24 mph constant speed, Efficiency of 84% for Torque of around 160 N-m and motor speed of 320 rpm for IWM. And as for CM, For 24 mph constant speed, CM has efficiency of around 70% (after considering transmission losses) for Torque of 20 N-m and motor speed of 3000 rpm.

All the experimental and the simulation results were obtained for two in-wheel motors whereas for getting the best efficiency and to optimize it accordingly, it is best to use four in-wheel motors which is the concept of low-level control to be used for Phase 2 of the project. Fig 2.6 and Fig 2.7 shows the Low Level control logic concept of using of two or four underpowered motors at the wheels instead of a single motor with a transmission^[12].

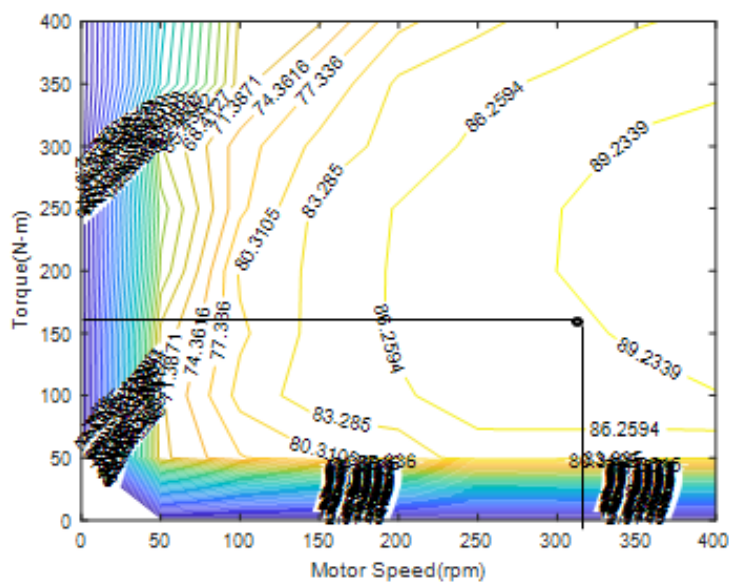


Fig 2.6: Low Level Control Logic with 2 IWMs

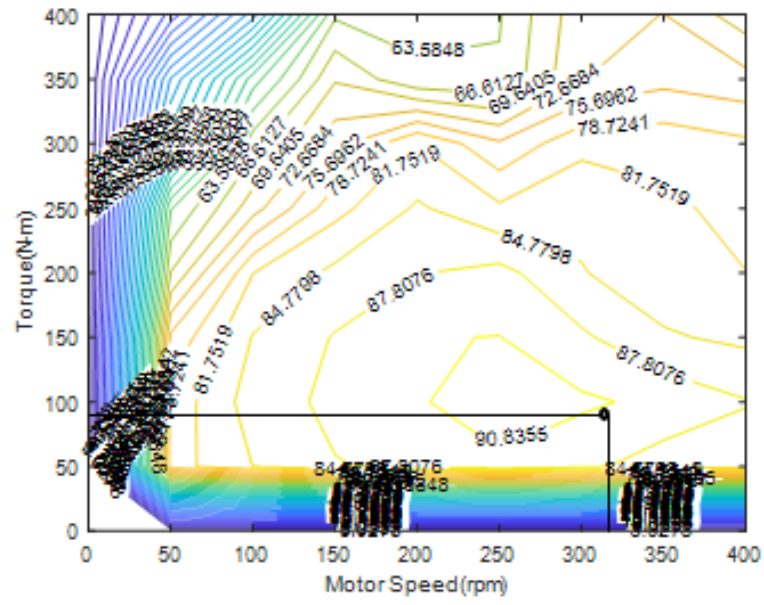


Fig 2.7: Low Level Control Logic with 4 IWMs

CHAPTER-3

HIGH LEVEL DECISION CONTROL AND RESULTS

High Level Control Concept:

The main concept and background behind High Level control is Energy efficient mobility (energy efficiency optimization for CAV Olli Shuttle with the Central & In-Wheel motor in urban city driving scenarios). The motivation behind this is the assumption that optimizing the speed trajectories Olli Shuttle CAV in urban city route will result in significant energy efficiency improvement. The objective behind this High Level Control^[3] is utilizing the V2X capabilities of the Olli Shuttle in order to optimize its speed trajectories to reduce the total trip time and energy wastage observed at traffic light regulated intersections. This High level control is achieved by local optimization and road segmentation. It is also worth mentioning that the initial approach was to use global optimization however this was not simple enough and will prove to be computationally expensive when we consider real time implementability.

For the Phase-1 of this project, only the V2X capabilities of the Olli Shuttle are utilized in order to optimize its speed trajectories to reduce the total trip time and energy wastage observed at traffic light regulated intersections. The optimization here is being done by local optimization and road segmentation where the road is segmented between traffic lights (where D_{TLx} is the distance between traffic lights x and x-1 as indicated in Fig 3.1).

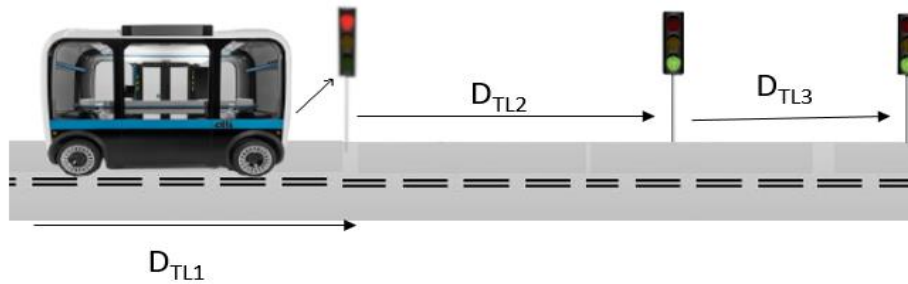


Figure. 3.1. Road Segmentation

Initial Drive Cycle & Assumptions:

The baseline drive cycle^[10] with the vehicle driven normally with traffic lights along the route is shown in Fig 3.2. This initial baseline drive cycle is then optimized in accordance with the high-level velocity optimization algorithm to get the optimized drive cycle. Here, the baseline drive cycle has stop signs along the route. However, for the scope of this project, the stop signs are neglected since the optimized drive cycle cannot avoid stoppage at the stop sign whatever

may be the velocity. The corresponding traffic lights corresponding to the distances are also shown underneath the figure in Fig 3.2.

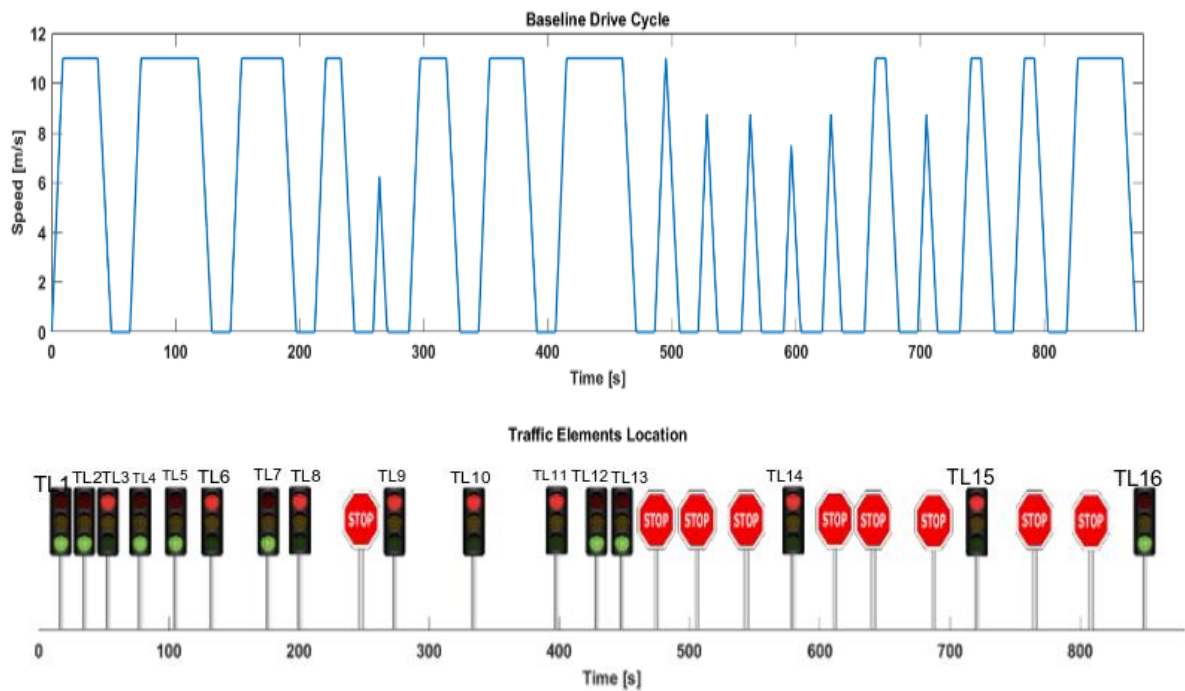


Fig 3.2: Baseline drive cycle with stop signs

As mentioned before, for the scope of this project, the Fig 3.2 with the baseline drive cycle route has been modified with no stop signs along the route. The modified baseline drive cycle with no stop signs is shown in Fig 3.3 below.

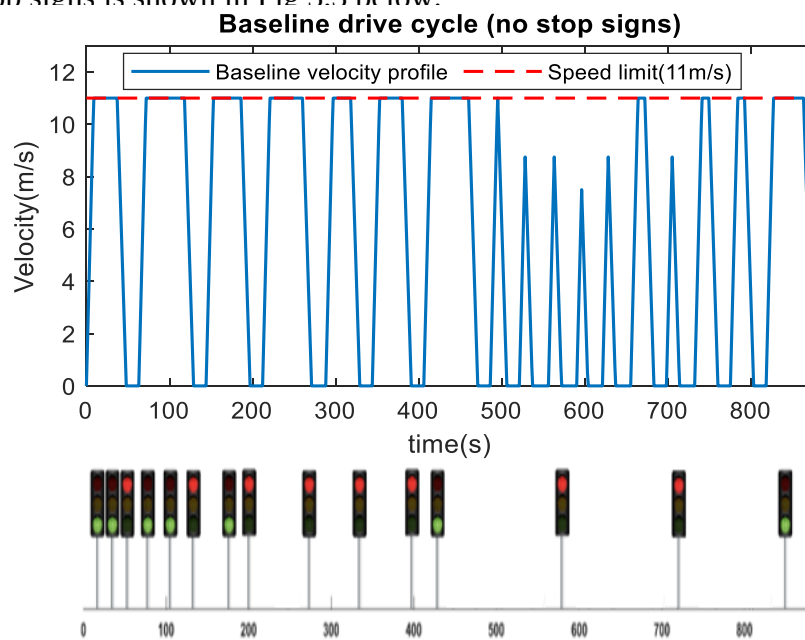


Fig 3.3: Baseline drive cycle with no stop signs

There are some assumptions for the baseline drive cycle which are as follows:

1. Traffic lights are synchronized with 30-second signal phase time (20s green, 10s red). YELLOW avoided (YELLOW same as green).
2. Olli accelerates from rest to Cruising speed with constant acceleration of 1.25 m/s^2 and decelerates from Cruising to rest with constant deceleration of 1.00 m/s^2 .
3. There are no STOP signs along the route.
4. All the traffic signals are initialized to all Green.

The consequent energy consumption due to the baseline drive cycle with no stop signs and no optimization is shown in Fig 3.4 below. The energy consumption observed for the entire drive cycle duration is around 0.18 kW-h.

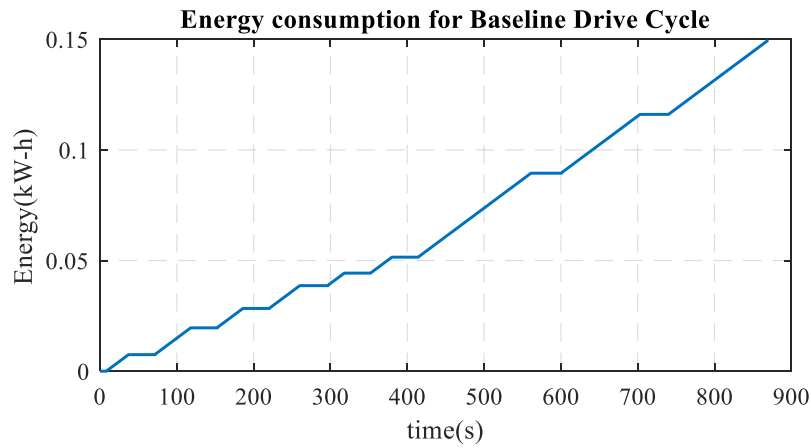


Fig 3.4: Energy consumption for baseline drive cycle

The First phase of optimization begins by finding the optimal speed trajectory to schedule a vehicle's arrival at a traffic light in order to eliminate or minimize the idling time and the consequent fuel wastage. The vehicle classifies the traffic features as either a stop sign or a traffic light. And, then the arrival time is calculated from the concept of $Time = \frac{Distance}{Speed}$

Velocity Optimization Algorithm:

From Fig 3.1, the distance between each traffic segment can be obtained by GPS which is then utilized to calculate the Time to Traffic Light (TTTL). This TTTL will give information on the phase of the Traffic Signal, i.e., whether it's red or green. Now, if the TTTL gives the signal as green, then the vehicle is going to keep moving at the same velocity since it can hit green. However, if the TTTL is red, then a new optimized velocity is calculated from the Optimization

Algorithm. In accordance with this algorithm, the vehicle is either going to speed or slow down to ensure minimum idling and energy wastage. The optimization algorithm is shown below:

$$\frac{TTTL_{TLx}}{SP} = n \frac{p}{q}$$

Where SP is the signal Phase and is equivalent to 30 seconds (20 seconds green(GP) + 10 seconds red(RP)). Dividing the TTTL by the SP is going to give the phase at which the vehicle is going to arrive (whether its red or green). $\frac{p}{q}$ here shows a fraction which refers to the time elapsed into the current phase (whether green or red) but which is just used to indicate the phase at which vehicle is arriving.

- $\frac{p}{q} < 0.667$ indicates arrival in green and $V_{optx} = V_{lim}$ (speed limit) or $V_{opt(x-1)}$
- $\frac{p}{q} > 0.667$ indicates arrival in red and $V_{optx} = \frac{D_{TLx}}{TTTL_{TLx} \pm WT}$

Where, WT is the waiting time and $WT = 30 * \left[1 - \left(\frac{p}{q}\right)\right]$. Here $\pm$ determined by $\frac{AT}{RP}$ where AT (called, Arrival Time) is the time elapsed into red phase RP.

Arrival time or Time elapsed at the current phase, $AT = 10 * [(3 * xy) - 2]$. Alternatively, logically, $AT = 10 - WT$.

This fraction is going to be used as a threshold to determine whether the optimum vehicle velocity is going to increase or decrease.

- When $\frac{AT}{RP} \geq 0.75$

$$V_{optx} = \frac{D_{TL1}}{TTTL_{TLx} + WT}$$

ie, the Olli slows down to avoid stoppage and waiting

- When $\frac{AT}{RP} \leq 0.25$

$$V_{optx} = \frac{D_{TL1}}{TTTL_{TLx} - WT}$$

ie, the Olli speeds up to ensure arrival at green and to avoid stoppage

When the vehicle arrival is in between the 2 thresholds shown above, then the Olli will wait at intersections. However, the energy wastage due to this is very small. And, it is not observed in the drive cycle described above.

Fig 3.5 belows shows the vehicle speed limit which is the maximum threshold for optimization

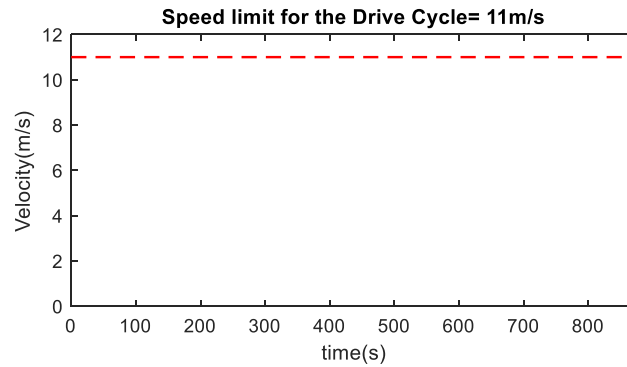


Fig 3.5: Vehicle maximum speed

Results:

The below Fig 3.6 shows the drive cycle with the optimized vehicle velocity.

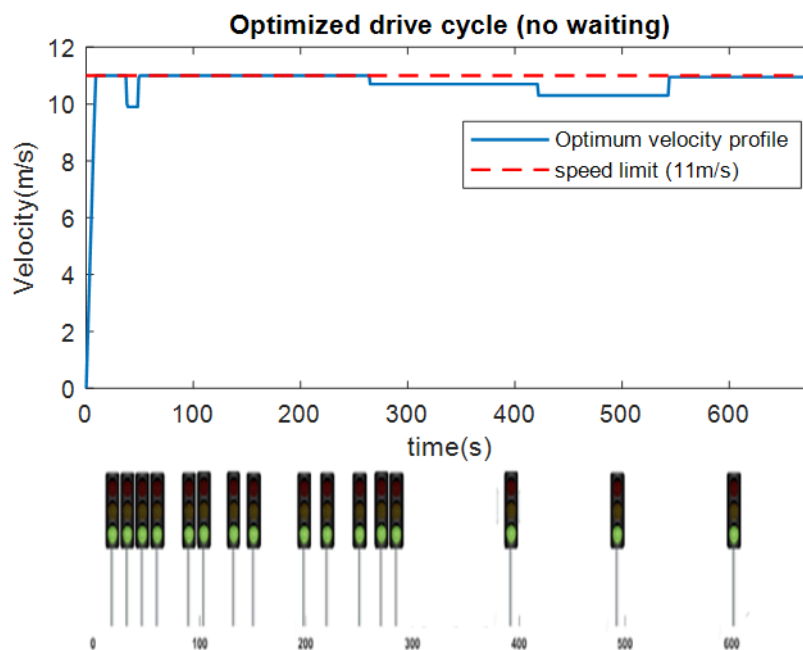


Fig 3.6: Optimized Vehicle velocity profile

Ideally, according to the algorithm, the optimum velocity can be either greater than or smaller than the speed limit. However, when the optimum velocity was obtained for our drive cycle, it was noted that there were no cases where the optimum velocity goes above the speed limit.

This velocity profile was then used to calculate the energy consumption and it is shown below in Fig 3.7.

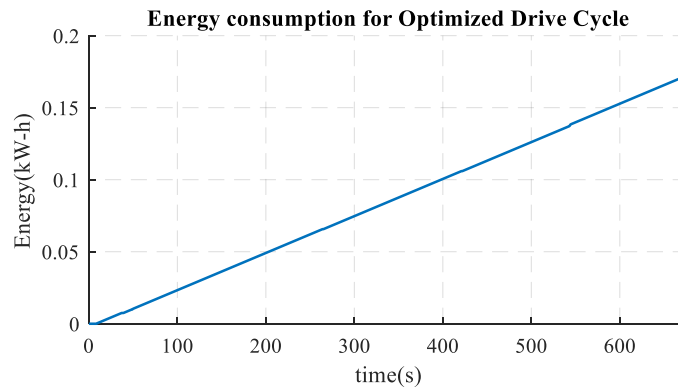


Fig 3.7: Energy consumption for the Optimized Drive Cycle

The above figure shows a reduction in energy consumption due to minimum idling of the optimized drive cycle and also a reduction in the total trip time. Here, it is observed that **there is about 10% energy savings & reduced trip time by about 18%.**

Next step, for the Phase 2 implementation, the algorithm is implemented along with the Intersection Manager algorithm in the Hardware in the Loop (HIL) before being tested in the vehicle. Afterwards, the high level controller is going to consider road network and optimize for route selection especially in order to avoid routes with Stop Signs and then integrating the same with the Intersection Manager.

Interface between High Level and Low Level:

As mentioned before, the high level controller gives inputs to the low level before the energy optimization takes place at the powertrain level. Additionally, the High Level looks after the velocity inputs along the route. However, there is an Intersection Manager (IM) which actually supersedes the High Level inputs at the intersections. Therefore, at the intersections, the IM takes over and gives the velocity inputs to the Low Level. Once the vehicle leaves the intersections, the High-Level controller takes over. Therefore, outside of intersections, there can be no velocity inputs coming from the IM, ie, $V_{IM}=0$.

The below flowchart shows the interaction between the High Level, Intersection Manager and the Low Level:

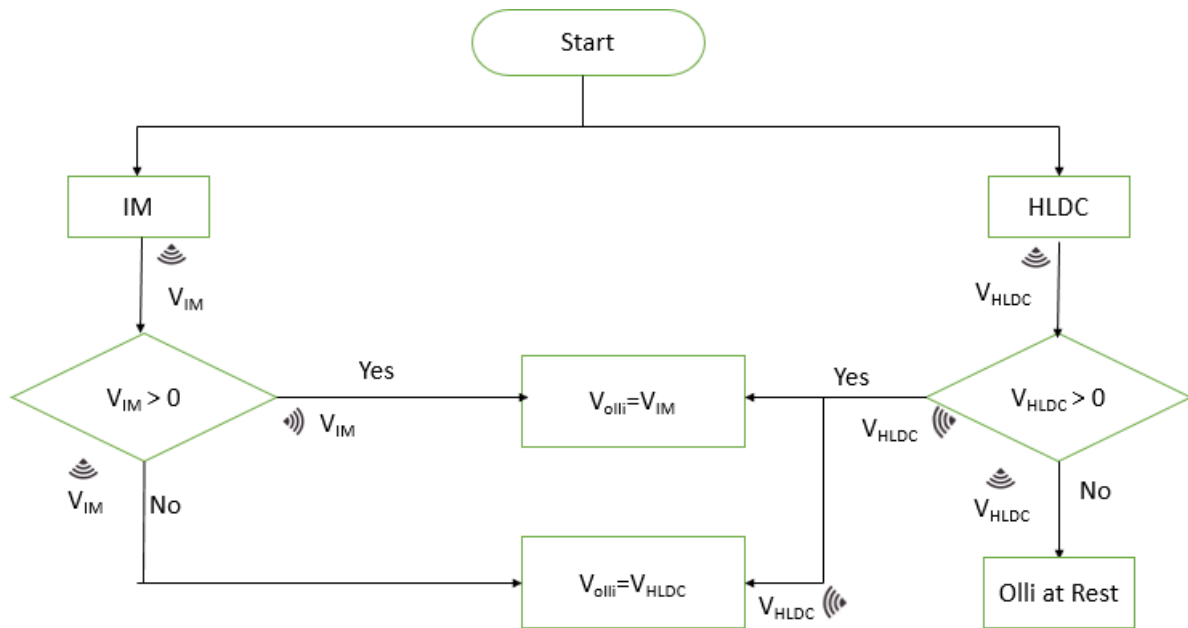


Fig 3.8: Interaction between the IM, High Level and the Low Level

In the flowchart, the V_{Olli} is presumed to the Olli vehicle velocity being fed to the Low Level for optimization.

CHAPTER-4

CONCLUSION AND FUTURE SCOPE

For the low level control, the simulation energy efficiency for EV driving showed power savings for IWM over the CM and similar results were obtained for the experimental energy efficiency analysis when the experimental run data were analyzed thus validating the simulation. The simulation results showed 8.8% power savings for IWM over CM whereas the experimental results for the similar velocity showed around 7.5% power savings for IWM over CM. The suggested operating method currently incorporates 2 IWMs. However, the Low Level Control Logic shows better savings if 4 IWMs were utilized for Energy Efficient Control Allocation (EECA)^{[4][13]}.

For the high level decision control, the results showed significant energy savings for the optimized route with the optimum velocity where the vehicle is not stopping at lights due to hitting the light when it turns green over the baseline route with regular stops along the route caused due to red lights. The final optimized route showed about 10% energy consumption savings over the baseline route and a reduced trip time by about 18%.

Future Work:

Because of the big energy savings observed in EVs and as a result of the energy deficit in the world, the autonomous EV research with Energy optimization at all levels is bound to increase exponentially.

For Phase 2 of the project, additional complexity to be added to provide further optimization opportunities:

- For Low Level powertrain control, Consider four-wheel steering and four in-wheel motors for energy optimization of lateral motion^[15].
- For High Level decision control, Consider route selection and other road users to provide path planning beyond the current speed profile.

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